Bharath Srinivas

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Bharath Srinivas | bharath2903 | US Citizen

EDUCATION

Master of science in Computer Science 05/2026
Illinois Institute of technology Chicago, U.S.A
- Concentrations: Data Analytics
- Related coursework: Computing in Python, Machine Learning, Big Data, Data Structures and Algorithms, Database Management

Bachelor of Technology in Computer Science 05/2023
Reva University Bengaluru, India

SKILLS

Soft Skills
Communication | Problem-Solving| Teamwork and Collaboration | Critical Thinking | Time Management

Technical skills
Languages: Python, SQL
ML/Analytics: XGBoost, scikit-learn, pandas, NumPy, Streamlit , Matplotlib, Seaborn, Excel
Databases: MySQL
Concepts: Statistical Inference, Hypothesis Testing, Regression, Classification

ACADEMIC PROJECTS

Credit Risk Classification Model 07/2025
Classification | Logistic Regression | Feature Engineering | Streamlit
- Developed a classification model to predict loan default risk using customer and bureau data, aiming to support lending decisions.
- Merged datasets, engineered new features (e.g., credit utilization, delinquency ratio), and selected Logistic Regression for high interpretability. Used Optuna for hyperparameter tuning.
- Achieved **accuracy of 0.93**, **recall of 0.94**, **ROC AUC of 0.98**, and **KS Statistic of 85%**. Deployed the solution as a Streamlit app.

Health Insurance Premium Predictor 06/2025
Regression | XGBoost | Streamlit | Python
- Built a model to predict health insurance premiums based on demographics and medical history to help insurers better estimate individual risk.
- Performed data segmentation based on age, engineered domain-specific features (e.g., normalized risk scores), and trained an XGBoost regression model.
- Achieved **R^2 of 0.98** and **RMSE of ~$ 1250**; segmentation reduced prediction errors for young users to less than **2%**. Deployed via Streamlit with real-time predictions.

Target Market Analysis and Hypotheses Testing for Credit Card Launch 01/2025
- Simulated a campaign to identify the best target segment for a credit card product for a fictional bank.
- Conducted EDA and hypothesis testing on 1,000+ customer records; designed and executed an A/B test on 100 users, using Z-tests and confidence intervals to validate campaign effectiveness.
- Observed a **conversion rate lift of 40%**, with a **statistically significant result (Z=2.75, p=0.003)** and an **increase in average transaction value of $236**.

PUBLICATIONS

Battery Management System(BMS80) to Improve Battery Life in Electric Vehicles 05/2023
Milestone Research Publications
- Co-authored research on optimizing lithium-ion battery lifespan by limiting charging to 80% SoC using **RESTful APIs** and **eco-charging algorithms**.
- Designed a microcontroller-based system to monitor battery metrics (SoC, temperature) and validated results with **real-world EV data**, reducing energy waste by **15%** and charging time by **30%**.